

DC-link capacitor sizing

75 V, 100 A_{rms} GaN phase leg (2 × EPC2361 per switch)

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Abstract

The DC-link capacitor of the symmetric EPC2361 phase-leg building block is sized from the switching-frequency current it must supply. RMS ripple currents are derived analytically and checked with a time-domain PWM model for a single leg and for three legs sharing one bus, over modulation index and power factor. The required effective capacitance is given versus allowed voltage ripple for 20–200 kHz. Finally, the ripple current is split between the individual capacitors using the Q3D-extracted copper of the building block. Main results: worst-case ripple current 50.0 A_{rms} for a leg on its own and 65.0 A_{rms} for a three-phase bus; at 50 kHz a $\pm 2\%$ (3 V pp) ripple needs 78 μ F effective per leg on a shared bus, which the selected $24 \times 10 \mu\text{F} + 12 \times 1 \mu\text{F}$ bank provides; each capacitor carries at most about 2 A_{rms}.

1 Assumptions and definitions

- $V_{\text{dc}} = 75 \text{ V}$, sinusoidal phase current $I = 100 \text{ A}_{\text{rms}}$ ($\hat{I} = 141 \text{ A}$), ideal switches, no dead time.
- Modulation: SVPWM (min–max zero sequence), common triangular carrier for all legs; analytical formulas are for SPWM.
- The DC source or bulk capacitor supplies the switching-period average of the input current; the DC-link capacitor supplies the rest (“HF” current). For a three-phase inverter the average input power is constant, so no low-frequency ripple appears.
- “Single leg”: a leg’s own capacitors supply its own HF current (blocks far apart, or a stand-alone leg). “Three legs”: three legs share one bank, so their HF currents partly cancel.
- Modulation index $M = \hat{V}_{\text{phase}}/(V_{\text{dc}}/2)$, $0 \dots 1.15$; power factor $\cos \varphi = 0 \dots 1$.

Script: `scripts/dclink_report.py` (all numbers and figures in this report).

2 RMS ripple current

With switching function $s_k \in \{0, 1\}$ and phase current i_k , the input current of leg k is $i_{\text{in},k} = s_k i_k$, with switching-period average $d_k i_k$. Its HF part has the period-mean square $d_k(1 - d_k) i_k^2$.

Single leg (SPWM). With $d = \frac{1}{2}(1 + M \sin \theta)$ and $i = \hat{I} \sin(\theta - \varphi)$, averaging over the fundamental:

$$I_{C,\text{leg}} = \frac{\hat{I}}{2} \sqrt{\frac{1}{2} - M^2 \left(\frac{1}{4} + \frac{\cos 2\varphi}{8} \right)} \quad (1)$$

It is largest at low M : $\hat{I}/(2\sqrt{2}) = 50.0 \text{ A}$ for 100 A_{rms}. At the rated point ($M = 1$, $\cos \varphi = 1$) it is 28.8 A.

Three legs on one bus (Kolar & Round).

$$I_{C,3\phi} = \hat{I} \sqrt{M \left[\frac{\sqrt{3}}{4\pi} + \cos^2 \varphi \left(\frac{\sqrt{3}}{\pi} - \frac{9M}{16} \right) \right]} \quad (2)$$

Maximum 65.0 A at $M \approx 0.60$, $\cos \varphi = 1.0$; 50.3 A at the rated point. This is the total for all three legs, about one third per leg.

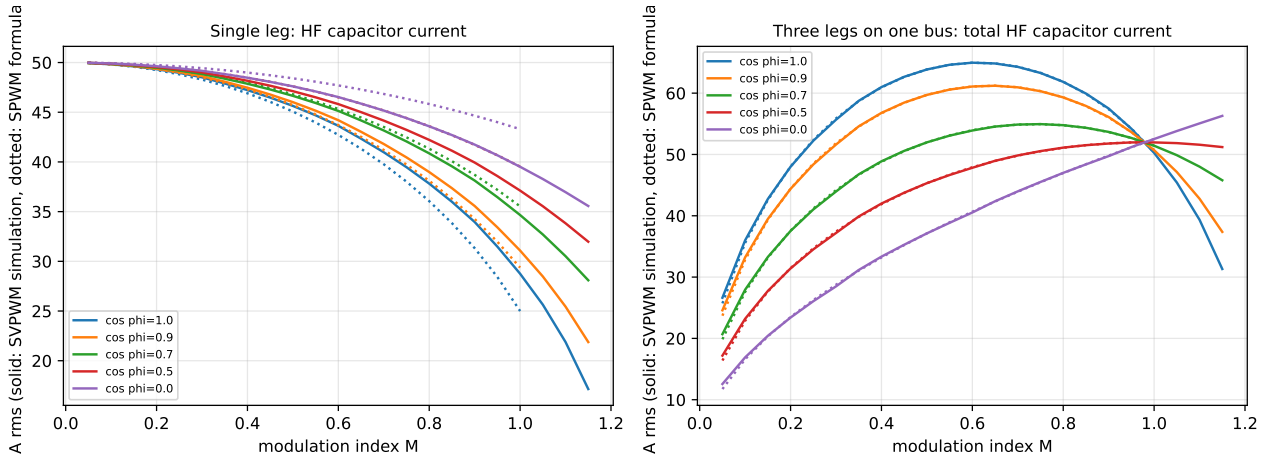


Figure 1: HF capacitor RMS current versus modulation index. Solid: time-domain SVPWM model; dotted: SPWM formulas above. They agree at low M ; above $M \approx 0.6$ the SVPWM zero sequence changes the duty cycles and the single-leg current is up to about 15% higher than the SPWM formula (28.8 A vs 25 A at $M = 1$). The three-leg total is almost unaffected.

3 Voltage ripple and required capacitance

For a capacitor C supplying the HF current, the worst peak-to-peak ripple within a switching period scales as $\Delta V_{pp} = K/(f_{sw}C)$.

- Single leg: the worst case is $d = 0.5$ at the current peak (low M), giving the hand formula $\Delta V_{pp} = \hat{I}/(4f_{sw}C)$. At 50 kHz this is 707 V for $1 \mu\text{F}$; the PWM model gives 707 V.
- Three legs on one bank: the PWM model gives 700 V for $1 \mu\text{F}$ of total capacitance at 50 kHz (worst over M and $\cos \varphi$). The total bank therefore needs about the same capacitance as one isolated leg, so each of the three blocks needs one third.

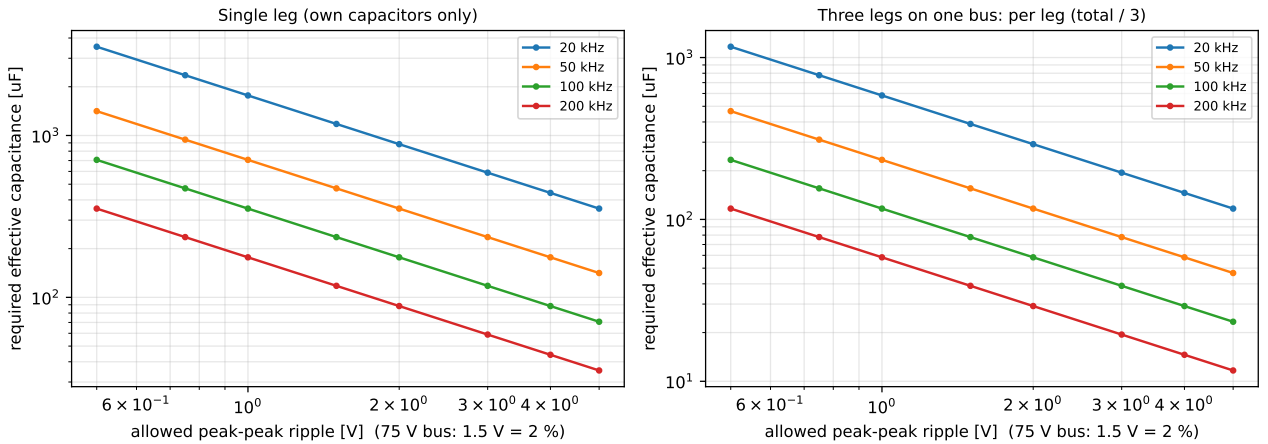


Figure 2: Required effective capacitance versus allowed peak-peak ripple. Left: single leg with its own capacitors. Right: three legs on one bus, per leg.

f_{sw}	0.75 V pp ($\pm 0.5\%$)		1.5 V pp ($\pm 1\%$)		3 V pp ($\pm 2\%$)	
	leg alone	per leg, shared	leg alone	per leg, shared	leg alone	per leg, shared
20 kHz	2357	778 (258)	1178	389 (128)	589	194 (64)
50 kHz	943	311 (102)	471	156 (51)	236	78 (25)
100 kHz	471	156 (51)	236	78 (25)	118	39 (12)
200 kHz	236	78 (25)	118	39 (12)	59	19 (5)

Table 1: Required effective capacitance in μF . In brackets: number of $10\text{ }\mu\text{F}/100\text{ V}$ 1210 MLCCs ($3\text{ }\mu\text{F}$ effective at 75 V) per leg, in addition to the 12 local $1\text{ }\mu\text{F}$ 0805 parts ($5.4\text{ }\mu\text{F}$ effective).

Selected bank per leg: $24 \times$ Murata GRM32EC72A106KE05L ($10\text{ }\mu\text{F}$, 100 V , X7S, 1210) and $12 \times$ TDK C2012X7S2A105K125AB ($1\text{ }\mu\text{F}$, 100 V , X7S, 0805). Assuming 30% and 45% of nominal at 75 V , this is $77\text{ }\mu\text{F}$ effective. That meets $\pm 2\%$ at 50 kHz and $\pm 1\%$ at 100 kHz on a shared three-phase bus ($78\text{ }\mu\text{F}$ required). A leg used on its own would need $471\text{ }\mu\text{F}$ for $\pm 1\%$ at 50 kHz ; that is a job for film or polymer capacitors on the DC bus, not for more MLCCs on the block. The DC-bias derating must be confirmed with the manufacturers' data (Murata SimSurfing, TDK SEAT).

Stand-alone single-phase half bridge

If one leg drives a load returned to the midpoint of a split DC link, each half of the link carries $i/2$ at the fundamental frequency f_1 . The midpoint ripple is then $\Delta V_{pp} \approx \hat{I}/(2\pi f_1 C_{half})$, which needs 150 mF per half at 50 Hz for 3 V pp , or 15.0 mF at 500 Hz . That is electrolytic-bank territory. It is the reason the block is intended for three-phase use, where this ripple cancels.

4 Capacitor RMS currents

The leg ripple current is not shared equally: the local 0805 banks sit inside the commutation loop, the 1210 rows are further away, and an external bus may take part of the low-frequency content. Method: the HF current of leg A from the PWM model is transformed (FFT over one fundamental period; bins holding 99.9% of the energy). Each harmonic is injected between the high-side drains and low-side sources of the Q3D building-block copper model (2 oz inner layers, inductance and resistance matrices). The copper resistance rises from the DC to the 100 MHz value as $\sqrt{f}$. Capacitors: 0805 $0.45\text{ }\mu\text{F}/6\text{ m}\Omega/0.48\text{ nH}$, 1210 $3\text{ }\mu\text{F}/3\text{ m}\Omega/0.5\text{ nH}$ each. Optional external bus: 20 nH and $2\text{ m}\Omega$ to a $1\text{ mF}/20\text{ m}\Omega$ bulk.

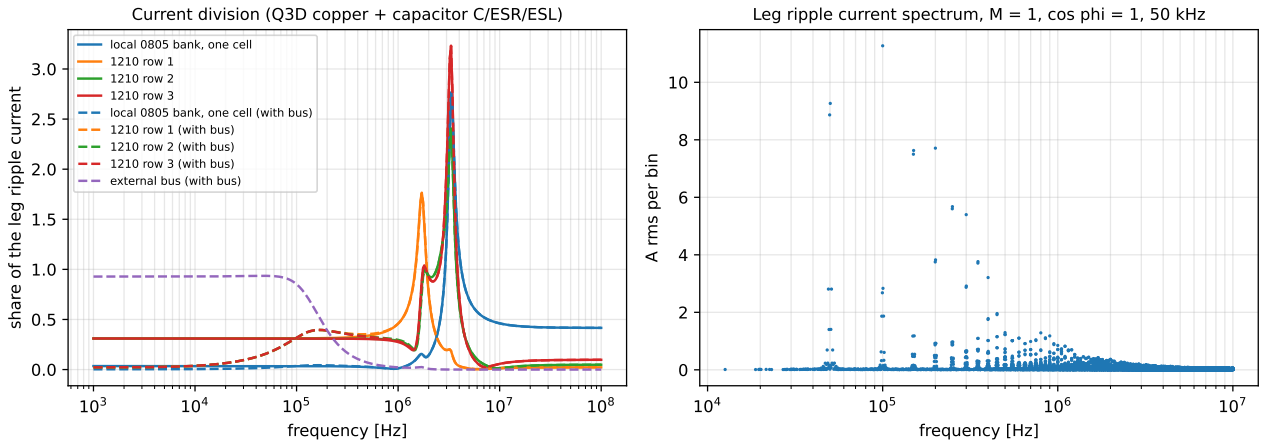


Figure 3: Left: fraction of a ripple-current harmonic carried by each capacitor group versus frequency (solid: block alone; dashed: with external bus). Right: ripple-current spectrum of one leg at the rated point.

RMS current per capacitor [A]	block alone		with external bus	
	rated (M=1, $\cos \varphi=1$)	worst (low M)	rated	worst
local 0805, $1\ \mu\text{F}$ (12 parts)	0.88	0.89	0.88	0.87
1210 row 1, $10\ \mu\text{F}$ (8 parts)	1.28	2.03	1.31	–
1210 row 2, $10\ \mu\text{F}$ (8 parts)	1.29	2.04	1.33	–
1210 row 3, $10\ \mu\text{F}$ (8 parts)	1.36	2.08	1.39	1.54
external bus (whole)	–	–	18.4	43.1
ESR loss, all on-board capacitors [W]	0.179	0.360	0.186	–

Observations:

- Each 1210 carries 1.3–2.1 A_{rms} and each 0805 about 0.9 A_{rms} . With the assumed ESR this is 5–13 mW per part, a self-heating of a few kelvin; confirm against the ripple-current rating in the manufacturer data.
- The external bus mainly takes the switching-frequency fundamental and its first harmonics. At 50 kHz the $77\ \mu\text{F}$ bank is about $40\ \text{m}\Omega$, comparable to a 1 mF bulk behind 20 nH, so a nearby bulk relieves the bank at low frequency. The on-board capacitors still carry the MHz content.
- The small local banks and the large 1210 rows form a resonant loop through the copper inductance, at about 1.8 MHz and 3.3 MHz. Near these frequencies current circulates between the groups at several times the injected value. The PWM spectrum carries little energy there, so the effect on RMS current is modest, but the resonance appears in the switching ringing and EMI. Damping options: one or two higher-ESR capacitors (or a series-R MLCC) across the bank, or fewer, larger local capacitors to move the resonance away.
- The current sharing depends on the assumed capacitor ESR and ESL. Use frequency-dependent manufacturer models (SimSurfing / SEAT S-parameters) for the final check.

5 Conclusions

- Ripple current: up to $50.0\ A_{\text{rms}}$ per leg if each leg is on its own, $65.0\ A_{\text{rms}}$ in total for three legs on a shared bus; 28.8 A / 50.3 A at the rated point.
- Capacitance: $\Delta V_{\text{pp}} \approx \hat{I}/(4fC)$ holds for a single leg and, with C the total bank, also for the three-leg worst case. Per leg on a shared bus: $156\ \mu\text{F}$ ($\pm 1\%$) or $78\ \mu\text{F}$ ($\pm 2\%$) at 50 kHz, half that at 100 kHz.
- The selected $24 \times 10\ \mu\text{F} + 12 \times 1\ \mu\text{F}$ bank ($77\ \mu\text{F}$ effective) meets $\pm 2\%$ at 50 kHz on a shared bus; each capacitor carries at most about $2\ A_{\text{rms}}$.
- A stand-alone single leg needs bus-level bulk capacitance (film or polymer for the HF ripple, electrolytic for any fundamental-frequency ripple).